UNIVERSITY OF MILANO-BICOCCA

Department of Informatics, Systems and Communication

Advanced Data Management

Project Report

City Mobility Platform — Data Management System

Academic Year 2025/2026

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Part 1: Relational vs Document Model

In this part, we build the same database using two different approaches: a traditional relational database (PostgreSQL) and a document database (MongoDB). We then compare how well each one handles our four queries at different data sizes.

1.1 Data Modeling

Relational Schema — PostgreSQL

In PostgreSQL, data is organized into four separate tables that are linked together using foreign keys (like references between tables):

- users — stores each user's name, surname, birthdate, and country
- stations — stores each station's name, city, and maximum vehicle capacity
- trips — stores each trip, linking a user to a start station and end station, with start/end times and cost
- events — stores each event that happened during a trip (GPS, ERROR, BATTERY, or DELAY)

Indexes were added on all linking columns (foreign keys) and on the event type column to make queries faster.

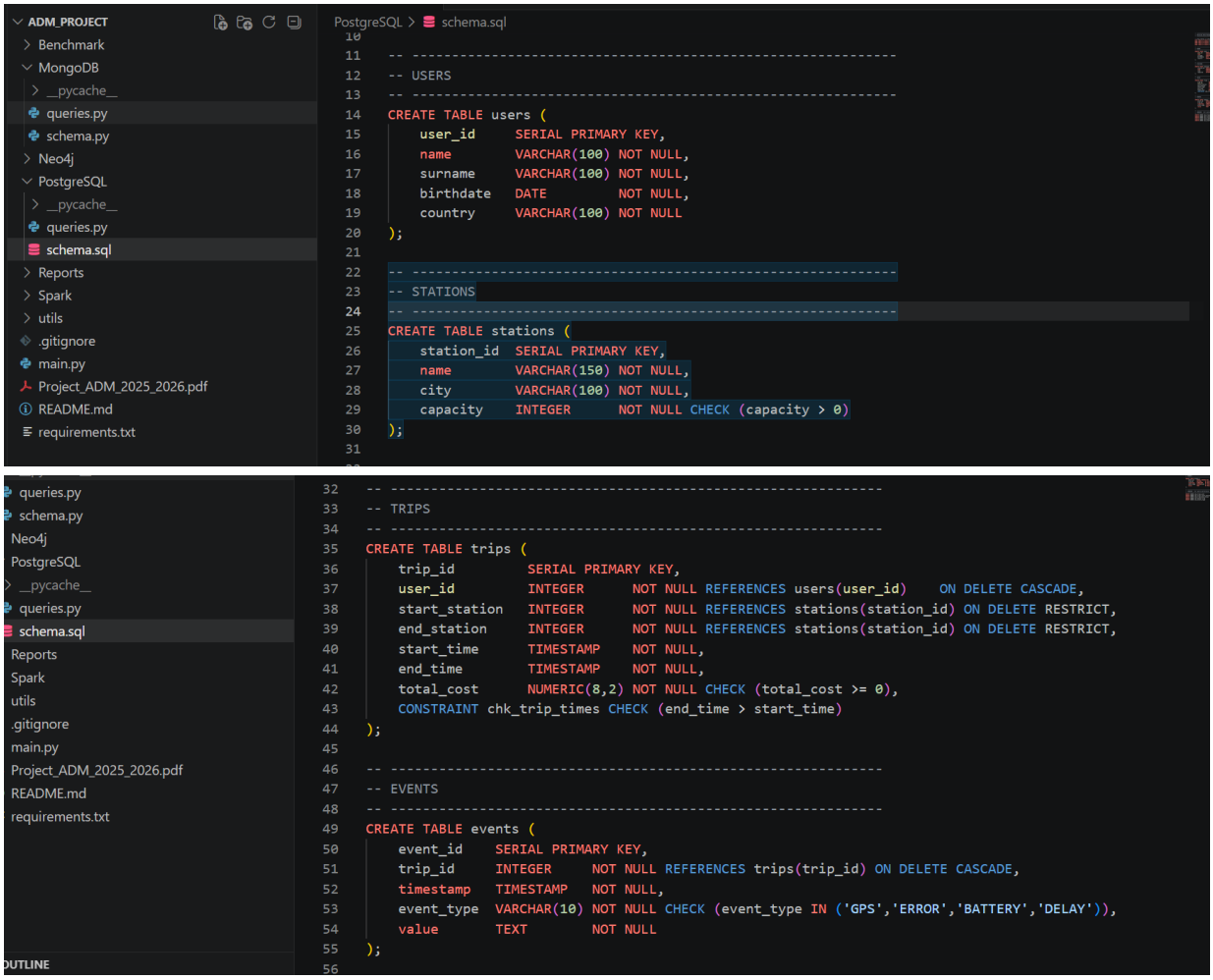


Figure 1: Relational Schema (PostgreSQL)

Document Schema — MongoDB

In MongoDB, data is stored as documents (similar to JSON files). We use three collections:

- users collection — one document per user
- stations collection — one document per station
- trips collection — one document per trip, with all events stored inside the trip document (embedded)

The most important design decision is that events are stored inside their trip document. This is called embedding. Users and stations are stored separately and referenced by ID. This is called referencing.

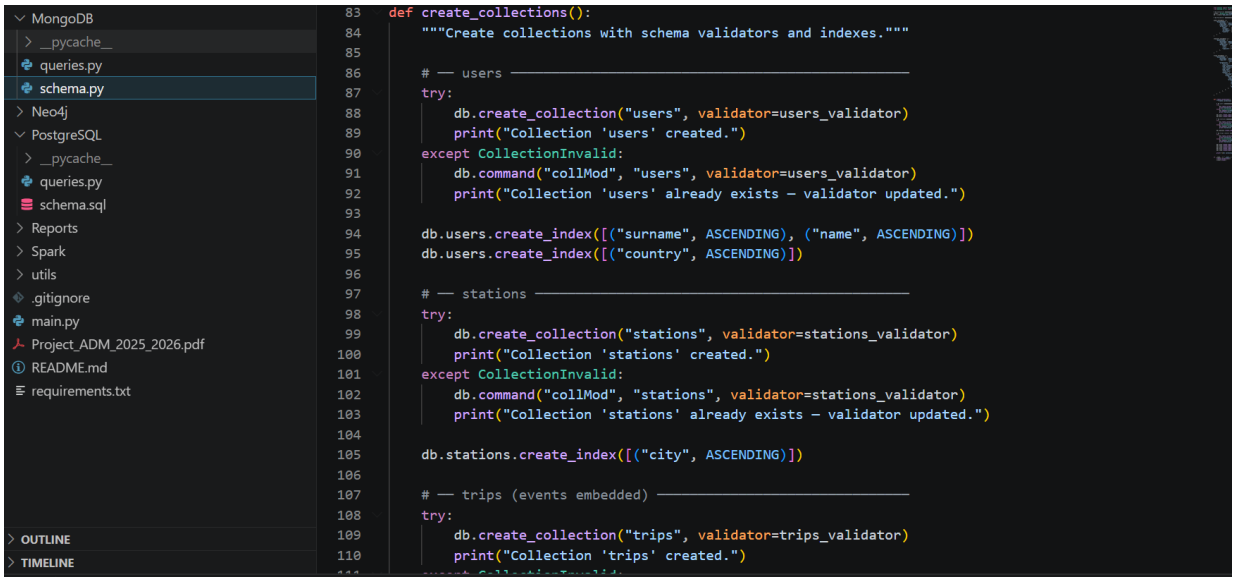


Figure 2: Document Schema (MongoDB)

Why Embedding vs Referencing?

What	Decision	Why
Events	EMBEDDED inside trips	Events only make sense as part of a trip. Storing them inside the trip avoids slow joins.
Users	REFERENCED (separate)	One user can have many trips. If we embedded user info in every trip, we would repeat the same data thousands of times.
Stations	REFERENCED (separate)	Same reason — only ~20 stations serve thousands of trips. Repeating station data would waste space.

1.2 Query Implementation

We implemented the same four queries in both PostgreSQL and MongoDB, then measured how long each one takes at different data sizes.

Query 1 — Show all trips with user details and station names

PostgreSQL Approach: We join three tables together — trips, users, and stations (twice for start and end). Because all linking columns are indexed, this is fast.

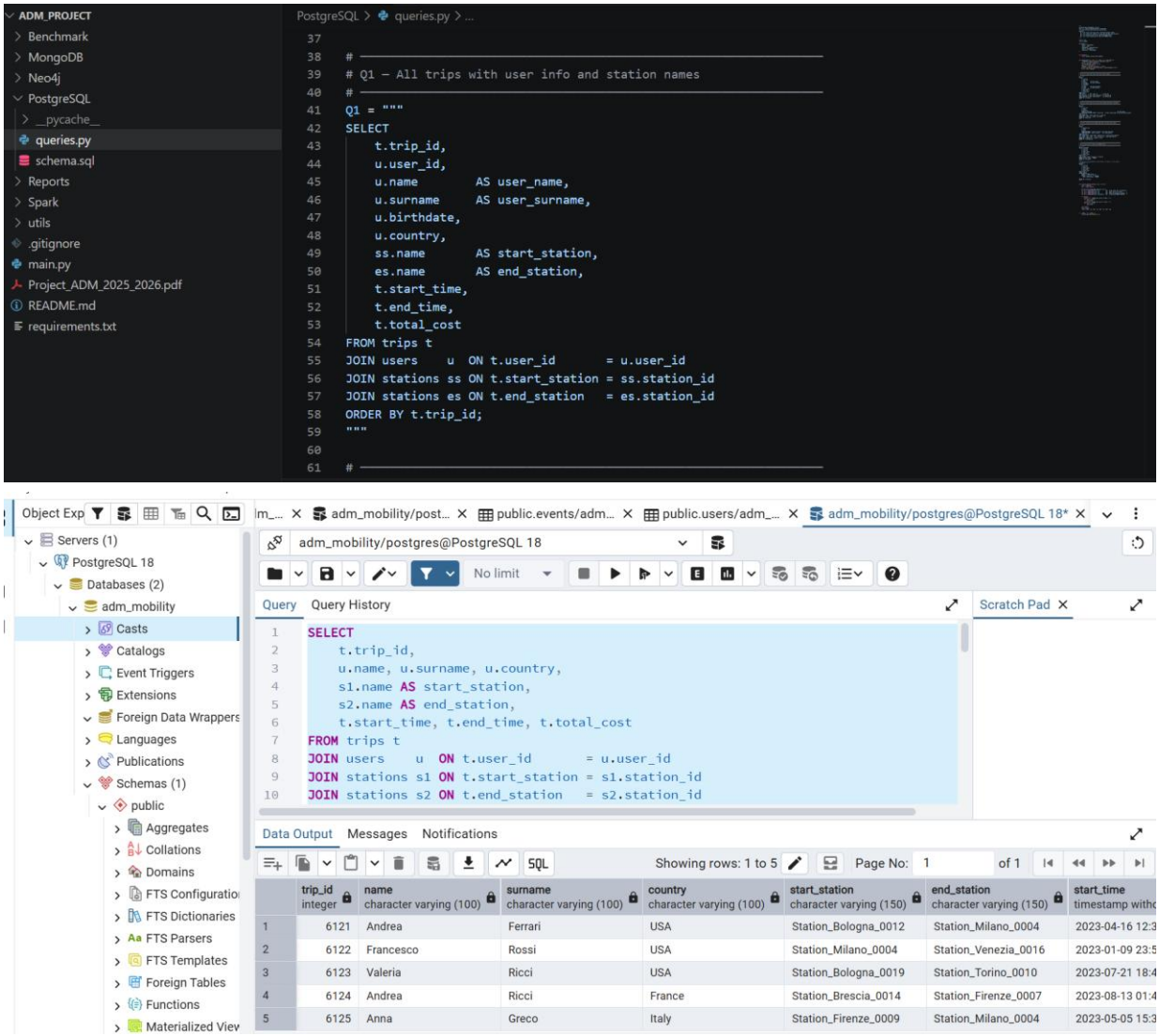


Figure 3: PostgreSQL - Q1: All trips with user information and station names

MongoDB Approach: We use three $lookup operations (similar to joins). This is slower than PostgreSQL because MongoDB has to look up referenced documents separately at query time.

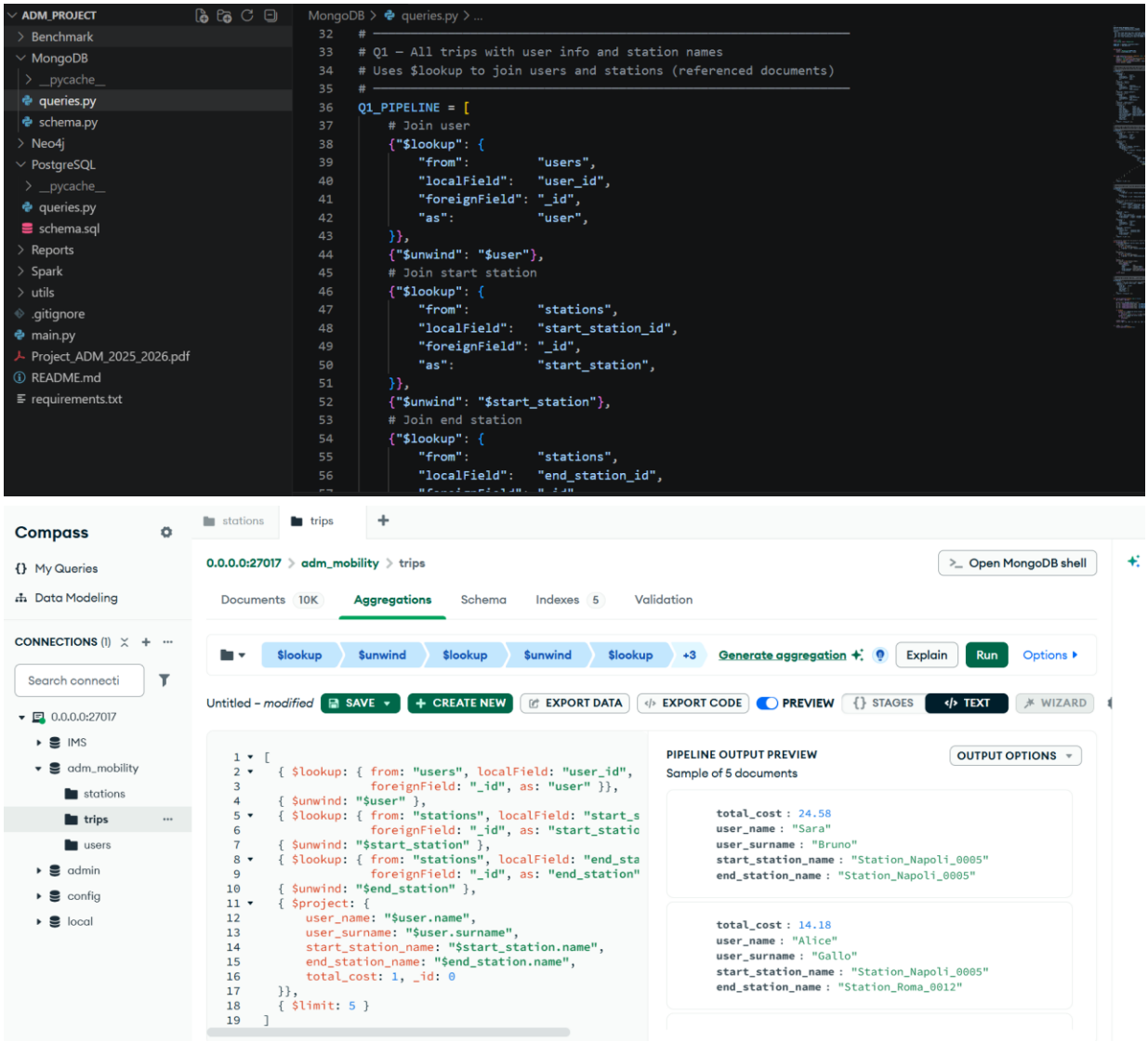
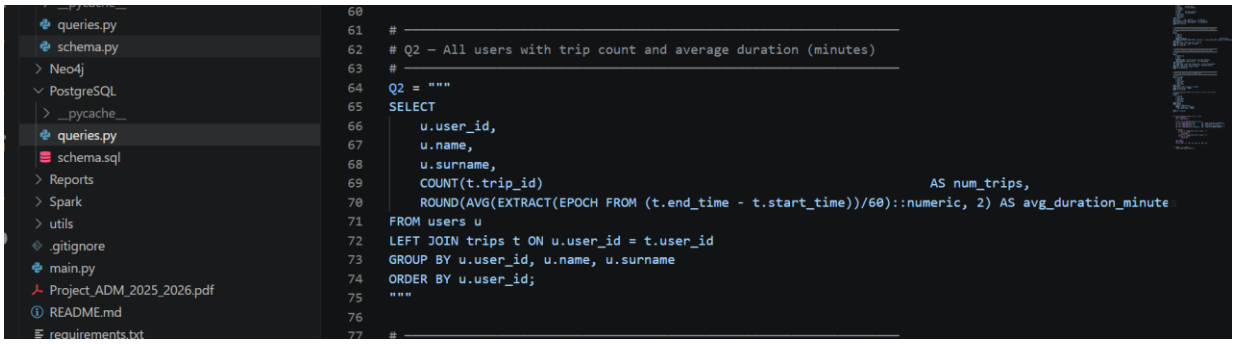


Figure 4: MongoDB - Q1: All trips with user information and station names

Query 2 — Show each user with their trip count and average trip duration

PostgreSQL Approach: Group all trips by user and calculate COUNT and AVG. Very efficient with indexed columns.



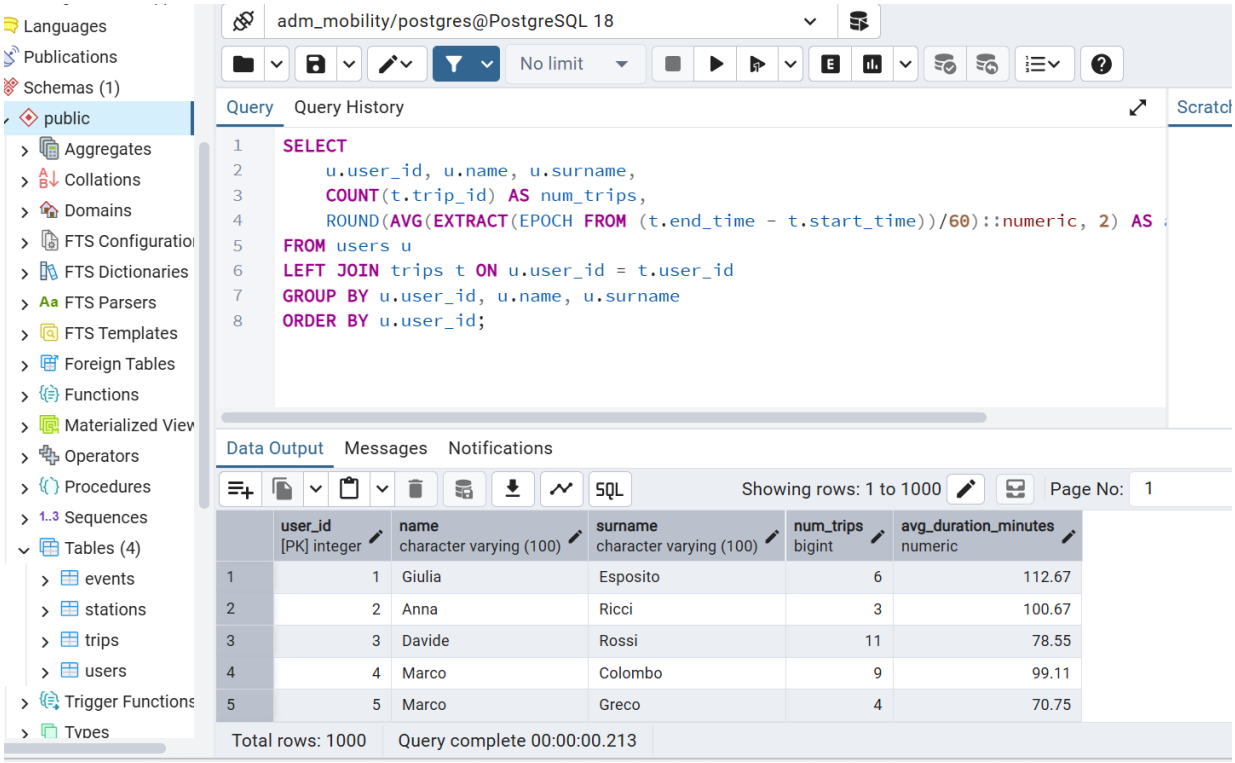
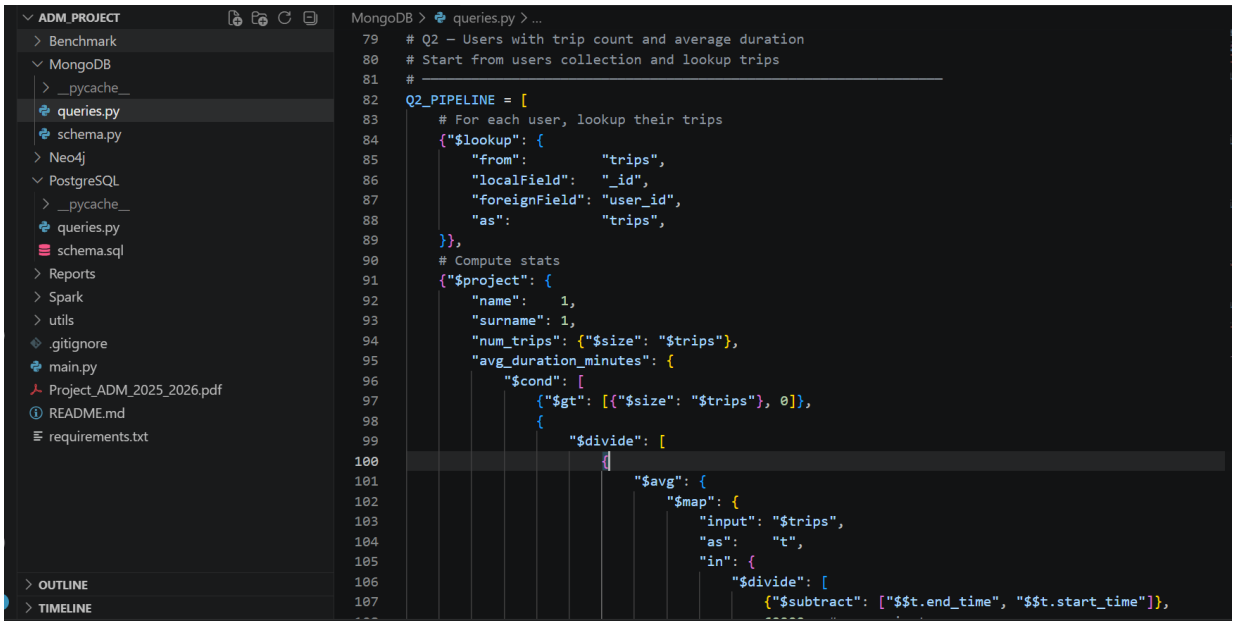


Figure 5: PostgreSQL - Q2: Users with number of trips and average trip duration

MongoDB Approach: Look up each user's trips and compute statistics. Slower when there are many users (50k) because of the lookup overhead.



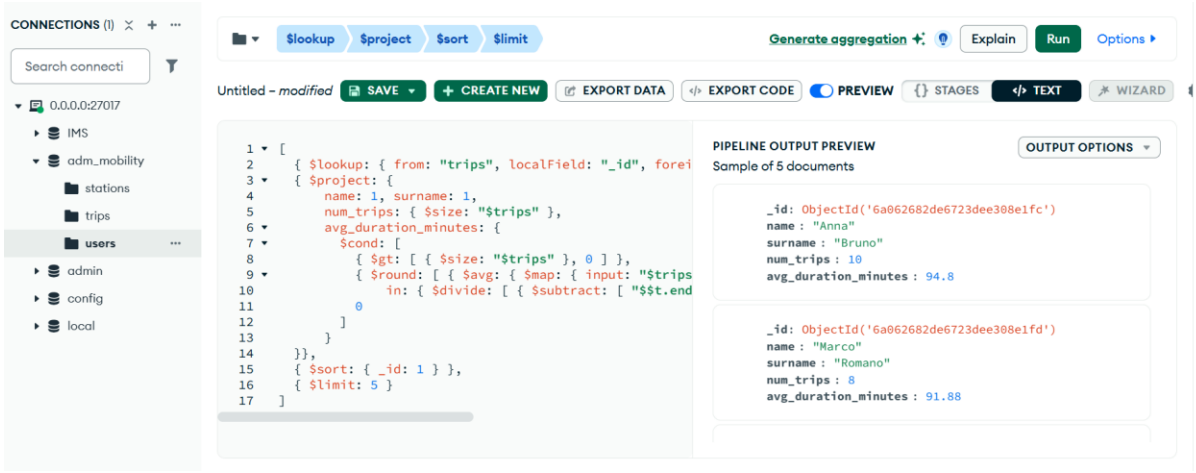


Figure 6: MongoDB - Q2: Users with number of trips and average trip duration

Query 3 — Show each station with how many trips start and end there

PostgreSQL Approach: Join the trips table twice (once for start, once for end). This creates a very large intermediate result, which makes it extremely slow at large scales.

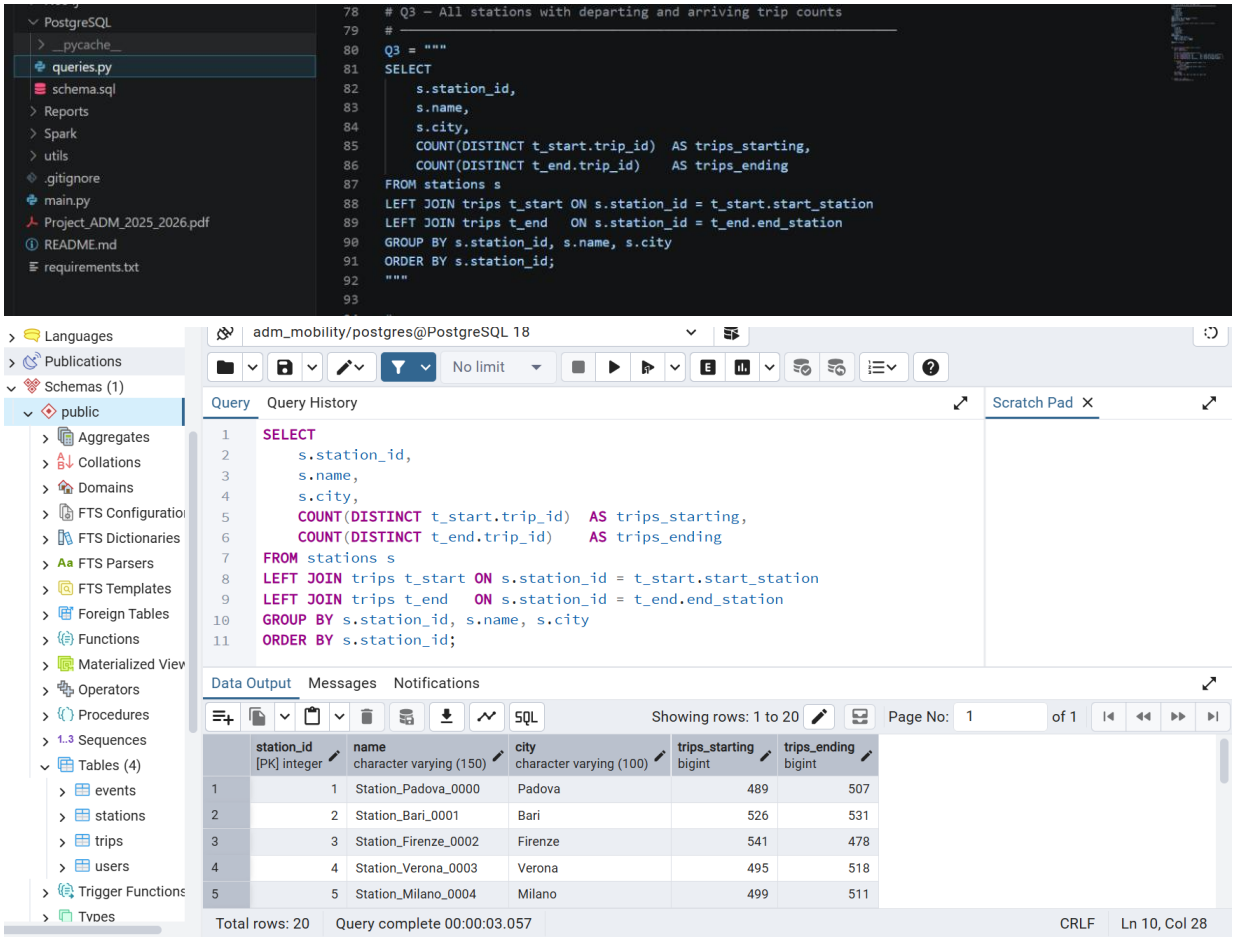


Figure 7: PostgreSQL - Q3: Stations with trips starting and ending there

MongoDB Approach: Use $facet to count start trips and end trips separately and efficiently. This is much, much faster than PostgreSQL at large scales.

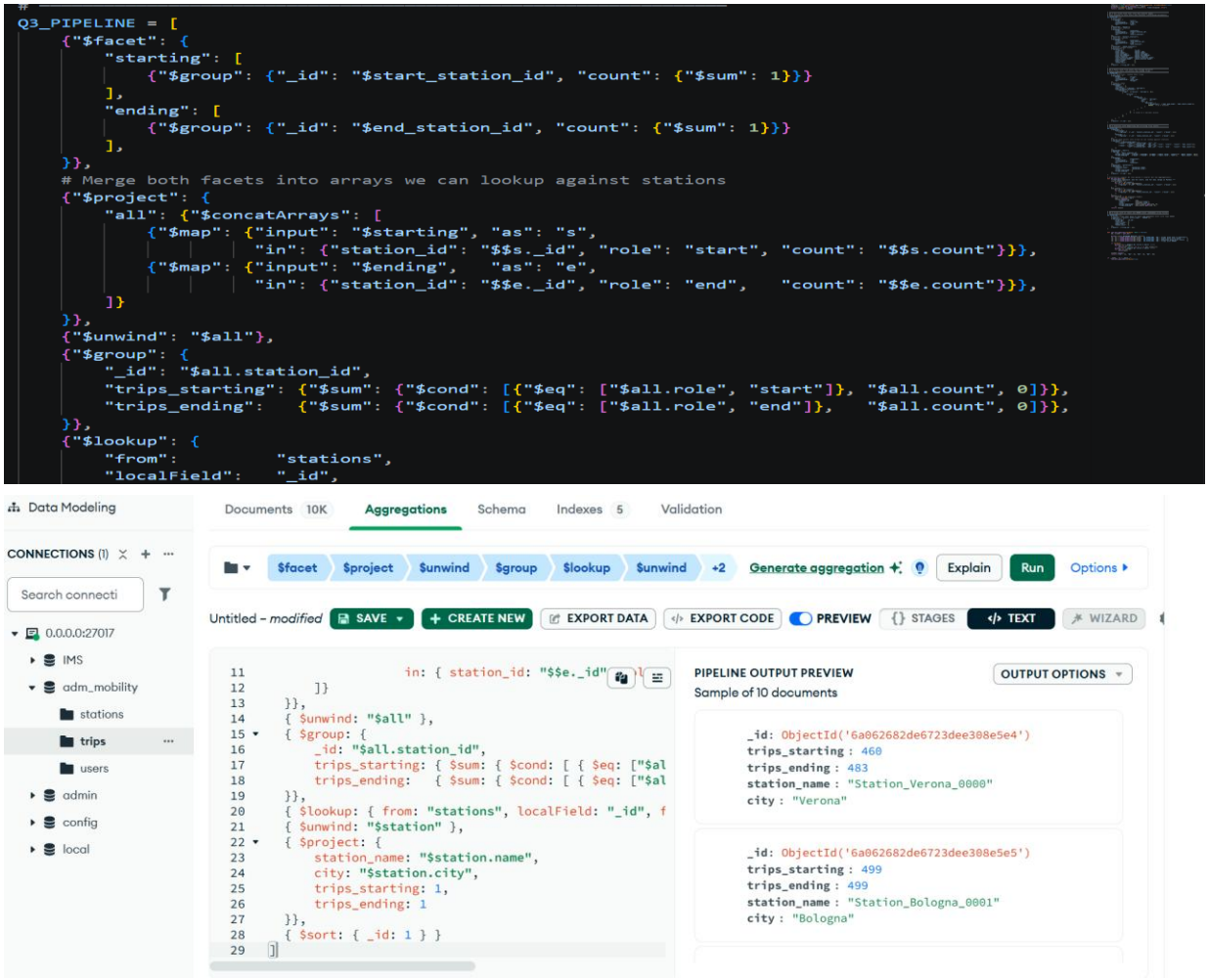
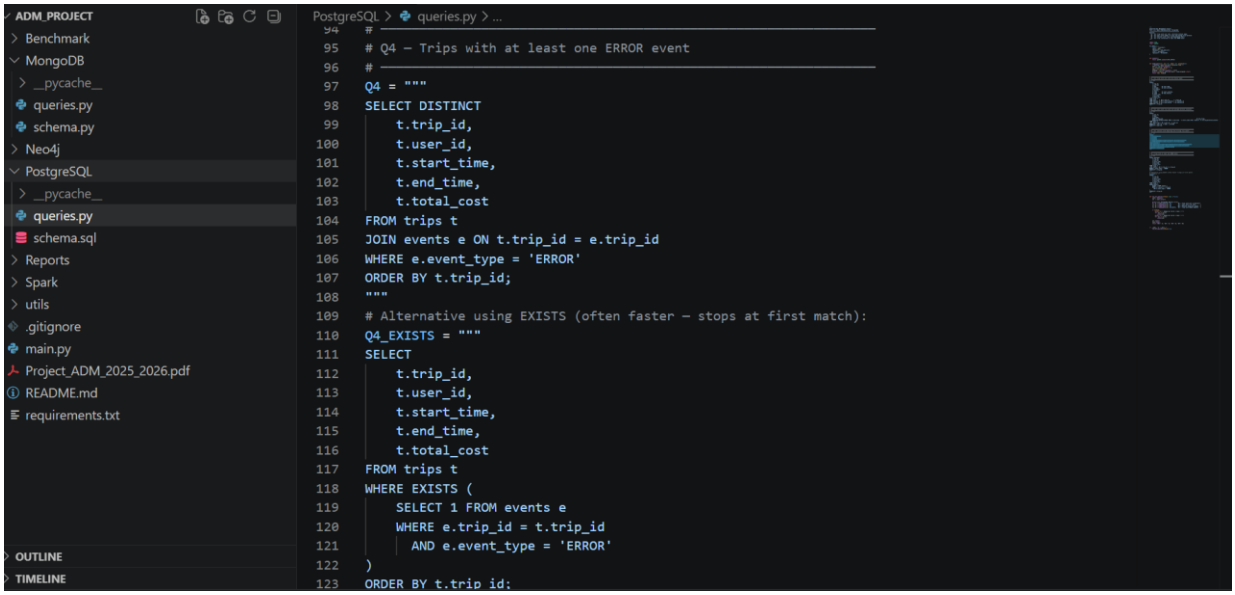


Figure 8: MongoDB -Q3: Stations with trips starting and ending there

Query 4 — Find all trips that have at least one ERROR event

PostgreSQL Approach: Use an EXISTS check on the events table — stops as soon as it finds one matching event, so it is efficient.



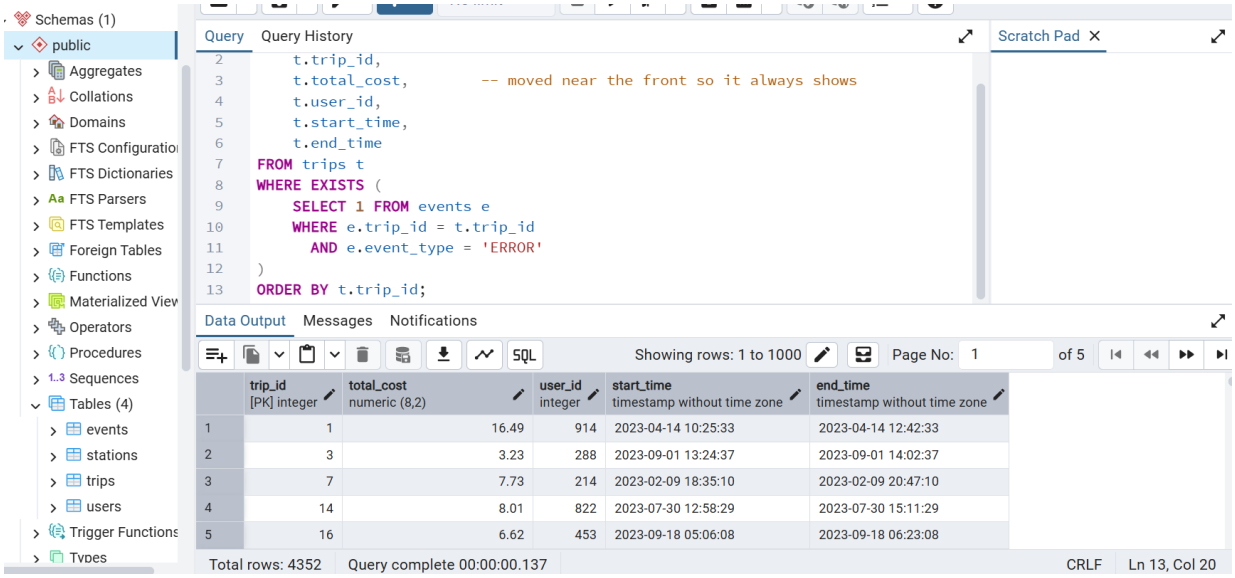


Figure 9: PostgreSQL - Q4: Trips containing at least one ERROR event

MongoDB Approach: Filter directly on the embedded events array. Because events are inside the trip document, no separate table lookup is needed.

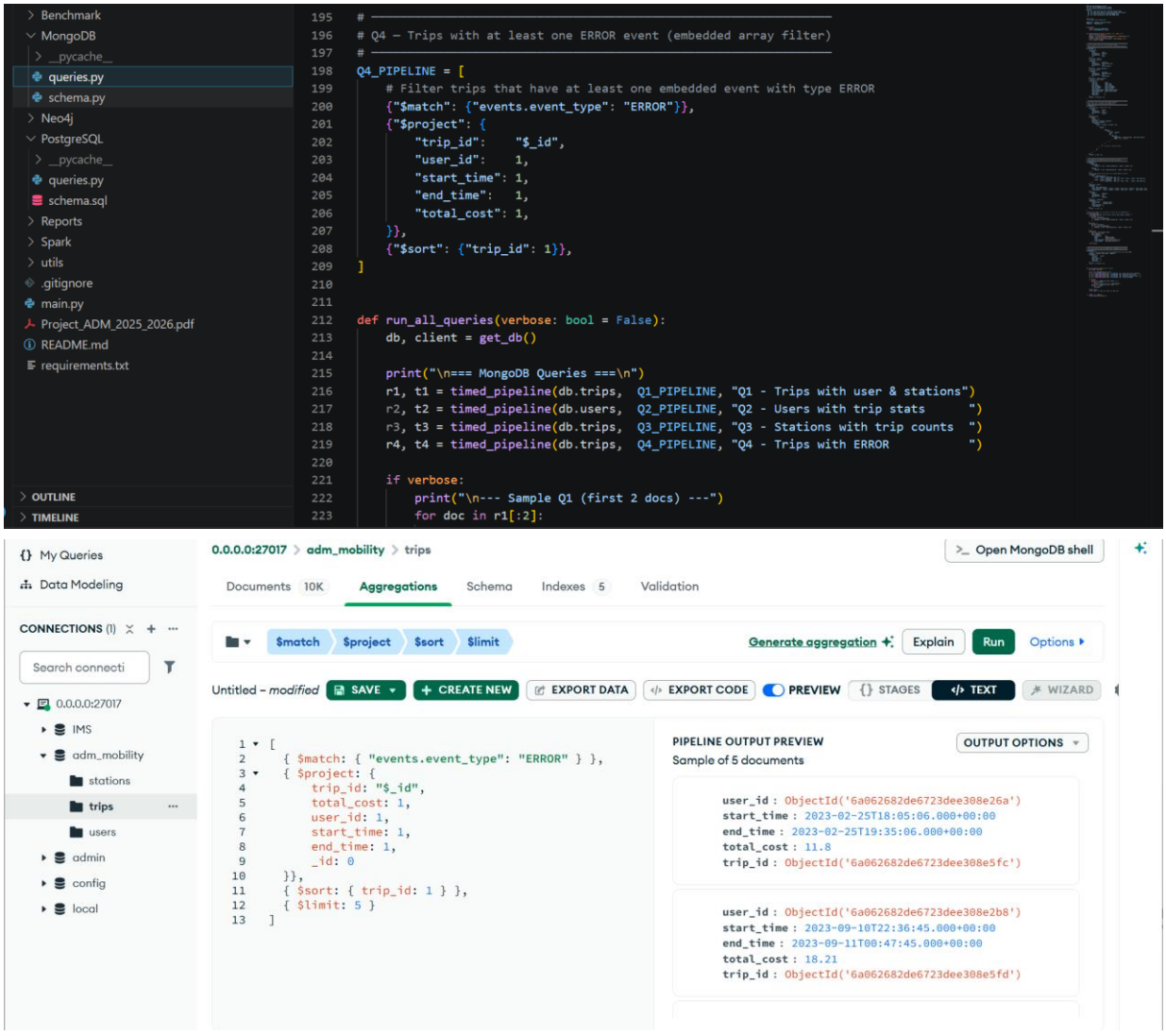


Figure 10: MongoDB - Q4: Trips containing at least one ERROR event

Benchmark Results — All 36 Scale Combinations

We tested every combination of: 1k/10k/50k users × 10k/50k/100k trips × 0/2/5/10 events per trip. Times shown in seconds (lower = faster):

DB	Users	Trips	Events	Q1 (s)	Q2 (s)	Q3 (s)	Q4 (s)
PG	1k	10k	0	0.20	0.02	3.92	0.02
MG	1k	10k	0	2.02	0.06	0.02	0.01
PG	1k	10k	2	0.50	0.01	3.26	0.16
MG	1k	10k	2	2.19	0.04	0.02	0.07
PG	1k	10k	5	0.58	0.02	11.71	0.64
MG	1k	10k	5	2.51	0.04	0.02	0.14
PG	1k	10k	10	0.78	0.04	5.18	0.45
MG	1k	10k	10	3.58	0.14	0.04	0.16
PG	1k	50k	0	3.08	0.03	146.71	0.01
MG	1k	50k	0	14.37	0.38	0.16	0.02
PG	1k	50k	2	2.94	0.07	206.13	1.08
MG	1k	50k	2	17.30	0.47	0.26	0.55
PG	1k	50k	5	5.26	0.08	228.59	2.10
MG	1k	50k	5	23.36	0.55	0.15	0.79
PG	1k	50k	10	3.35	0.11	214.55	2.32
MG	1k	50k	10	15.28	0.34	0.23	0.59
PG	1k	100k	0	5.21	0.12	957.68	0.01
MG	1k	100k	0	21.67	0.49	0.30	0.02
PG	1k	100k	2	6.13	0.09	721.25	1.59
MG	1k	100k	2	29.55	0.99	0.42	0.68
PG	1k	100k	5	5.91	0.06	770.31	1.33
MG	1k	100k	5	21.98	0.86	0.23	0.95
PG	1k	100k	10	2.18	0.05	751.89	1.34
MG	1k	100k	10	32.39	1.04	0.39	1.35
PG	10k	10k	0	0.23	0.04	0.19	0.09
MG	10k	10k	0	2.55	0.25	0.05	0.02
PG	10k	10k	2	0.42	0.06	0.30	0.10
MG	10k	10k	2	3.63	0.37	0.08	0.10
PG	10k	10k	5	0.41	0.04	0.29	0.37

DB	Users	Trips	Events	Q1 (s)	Q2 (s)	Q3 (s)	Q4 (s)
MG	10k	10k	5	4.58	0.49	0.04	0.19
PG	10k	10k	10	0.35	0.07	0.21	0.19
MG	10k	10k	10	3.50	0.52	0.11	0.20
PG	10k	50k	0	1.65	0.08	19.02	0.01
MG	10k	50k	0	9.89	0.45	0.15	0.02
PG	10k	50k	2	1.43	0.15	20.73	0.51
MG	10k	50k	2	16.44	0.56	0.17	0.28
PG	10k	50k	5	1.29	0.05	11.89	0.57
MG	10k	50k	5	9.76	0.68	0.12	0.48
PG	10k	50k	10	1.10	0.08	11.61	0.72
MG	10k	50k	10	9.58	0.49	0.09	0.42
PG	10k	100k	0	1.91	0.08	60.55	0.00
MG	10k	100k	0	27.83	0.85	0.32	0.05
PG	10k	100k	2	2.03	0.11	51.28	0.68
MG	10k	100k	2	25.27	0.77	0.32	0.71
PG	10k	100k	5	2.15	0.09	55.11	1.13
MG	10k	100k	5	25.29	0.94	0.29	1.02
PG	10k	100k	10	2.22	0.24	58.63	1.34
MG	10k	100k	10	20.86	0.76	0.27	0.93
PG	50k	10k	0	0.31	0.23	0.08	0.07
MG	50k	10k	0	2.04	0.89	0.12	0.02
PG	50k	10k	2	0.32	0.29	0.04	0.06
MG	50k	10k	2	2.31	1.12	0.04	0.05
PG	50k	10k	5	0.26	0.14	0.04	0.11
MG	50k	10k	5	2.24	1.39	0.13	0.12
PG	50k	10k	10	0.30	0.12	0.04	0.16
MG	50k	10k	10	2.34	1.15	0.03	0.11
PG	50k	50k	0	1.12	0.14	0.71	0.00
MG	50k	50k	0	11.95	1.66	0.18	0.03
PG	50k	50k	2	1.16	0.17	0.79	0.29
MG	50k	50k	2	10.08	1.34	0.15	0.28
PG	50k	50k	5	1.13	0.16	0.63	0.52
MG	50k	50k	5	12.41	1.88	0.18	0.50

DB	Users	Trips	Events	Q1 (s)	Q2 (s)	Q3 (s)	Q4 (s)
PG	50k	50k	10	1.12	0.17	0.61	0.61
MG	50k	50k	10	12.69	1.65	0.17	0.52
PG	50k	100k	0	2.28	0.34	7.39	0.00
MG	50k	100k	0	27.18	2.11	0.34	0.02
PG	50k	100k	2	2.21	0.19	5.42	0.63
MG	50k	100k	2	22.61	2.01	0.41	0.42
PG	50k	100k	5	2.18	0.15	4.47	0.98
MG	50k	100k	5	22.57	2.14	0.56	1.07
PG	50k	100k	10	2.09	0.18	4.93	1.27
MG	50k	100k	10	20.36	1.72	0.29	0.93

What the Results Tell Us

Query 1 (trips with user and station info): PostgreSQL is clearly faster. At 50k users and 100k trips, PostgreSQL takes 2.09 seconds while MongoDB takes 20.36 seconds. SQL joins on indexed columns are very efficient.

Query 2 (user statistics): PostgreSQL wins again, especially when there are many users. At 50k users and 100k trips, PostgreSQL takes 0.18 seconds vs MongoDB's 1.72 seconds. MongoDB slows down because it must look up each user's trips separately.

Query 3 (station trip counts): MongoDB wins dramatically. The most extreme example: at 1k users and 100k trips, PostgreSQL takes 957 seconds (about 16 minutes!) while MongoDB takes only 0.30 seconds. This is because PostgreSQL's double join creates an enormous temporary result that is very slow to process.

Query 4 (trips with ERROR events): Results depend on event density. With no events, both are very fast. With 10 events per trip, both slow down similarly. MongoDB's embedded events give a slight advantage because no separate table lookup is needed.

1.3 Schema Evolution — Adding a New Field

What happens if we need to add a 'battery_level' field (a number from 0 to 100) to all BATTERY events? Here is how each database handles this change:

PostgreSQL — How to add the field

We need to run this command: ALTER TABLE events ADD COLUMN battery_level INTEGER CHECK (battery_level BETWEEN 0 AND 100);

This adds the column to every existing row with a NULL value (meaning 'no data'). For very large tables, this command can lock the table temporarily, which means no one can use it while the change is being made. The application code also needs to be updated to fill in battery_level for BATTERY events.

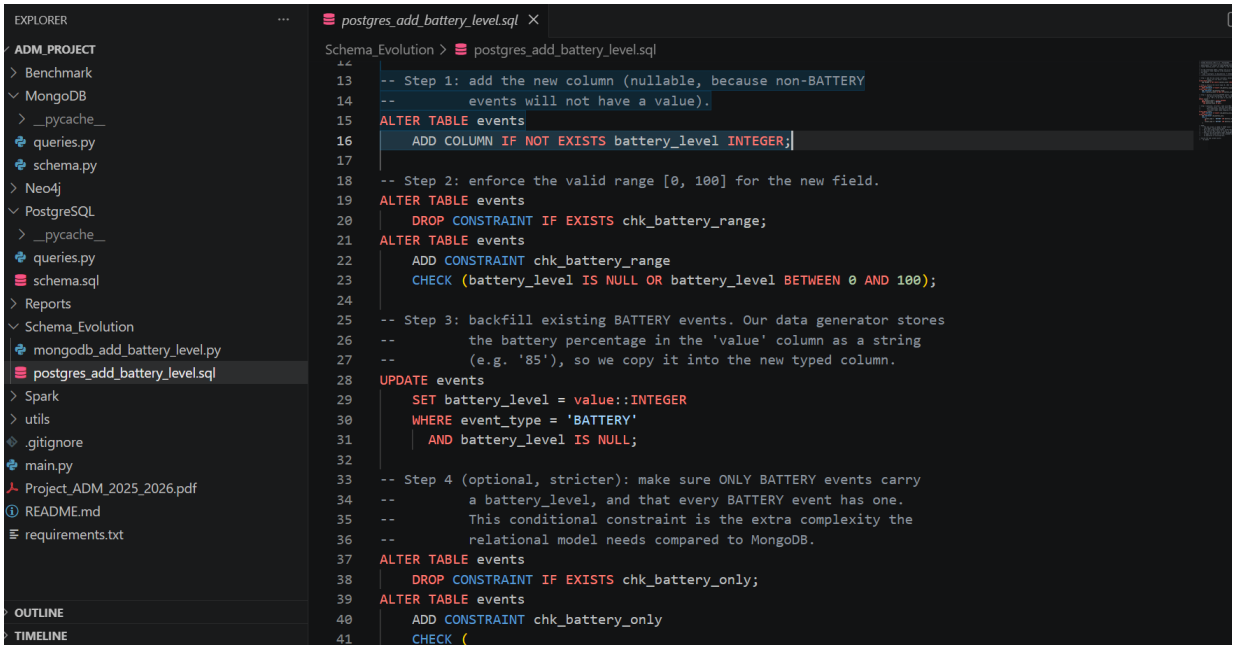


Figure11: PostgreSQL Schema Evolution ALTER TABLE migration adding battery level to BATTERY events

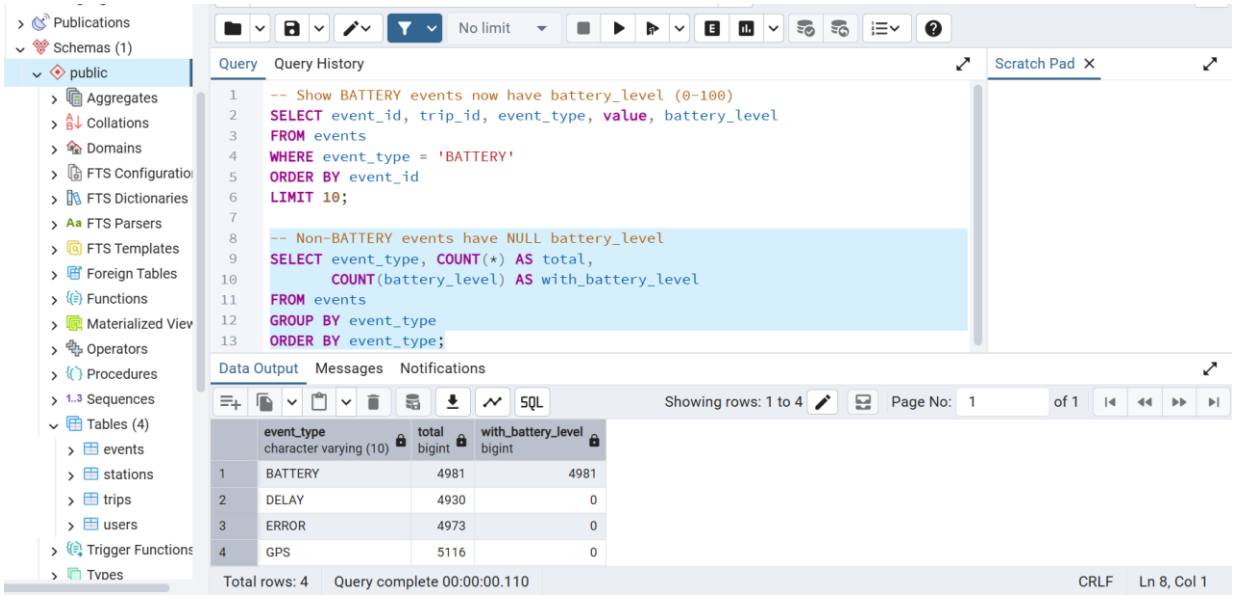


Figure12: PostgreSQL — BATTERY events with the new battery_level field populated

MongoDB — How to add the field

In MongoDB, we do not need to run any migration command at all. We simply start saving new BATTERY event documents with the battery_level field included: { event_type: 'BATTERY', value: '85', battery_level: 85 }. Old documents without the field remain perfectly valid. MongoDB is flexible enough to handle documents with different fields in the same collection.

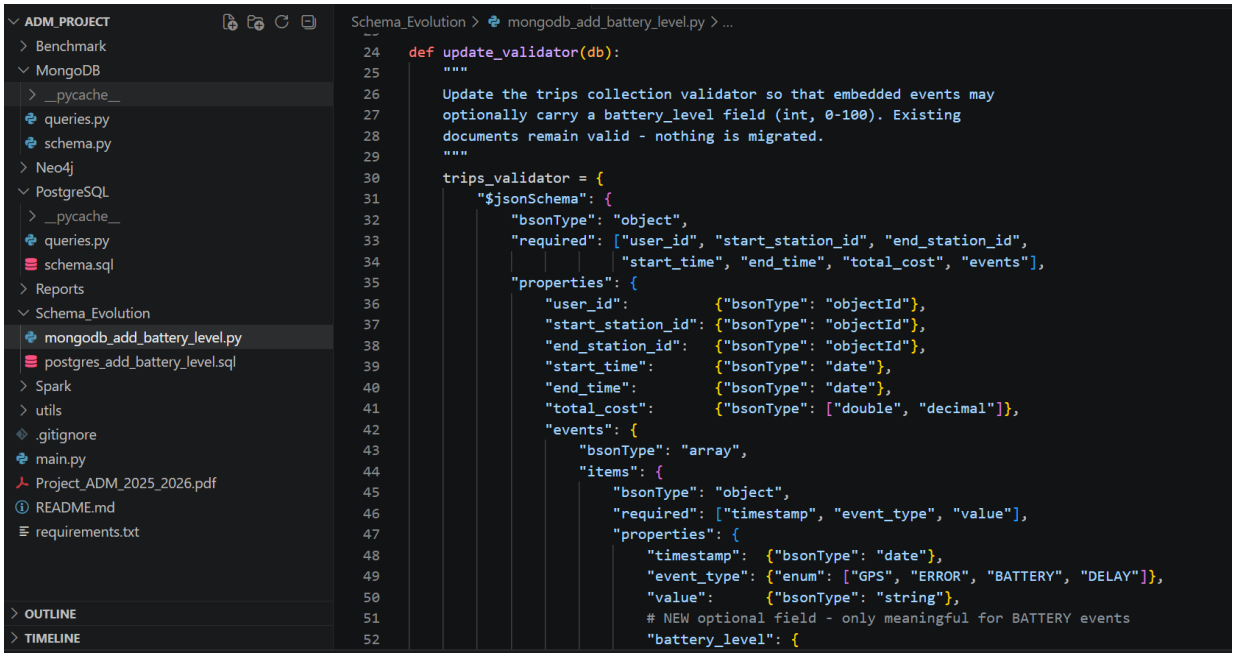


Figure13: MongoDB Schema Evolution -adding battery_level requires only one line in the validator

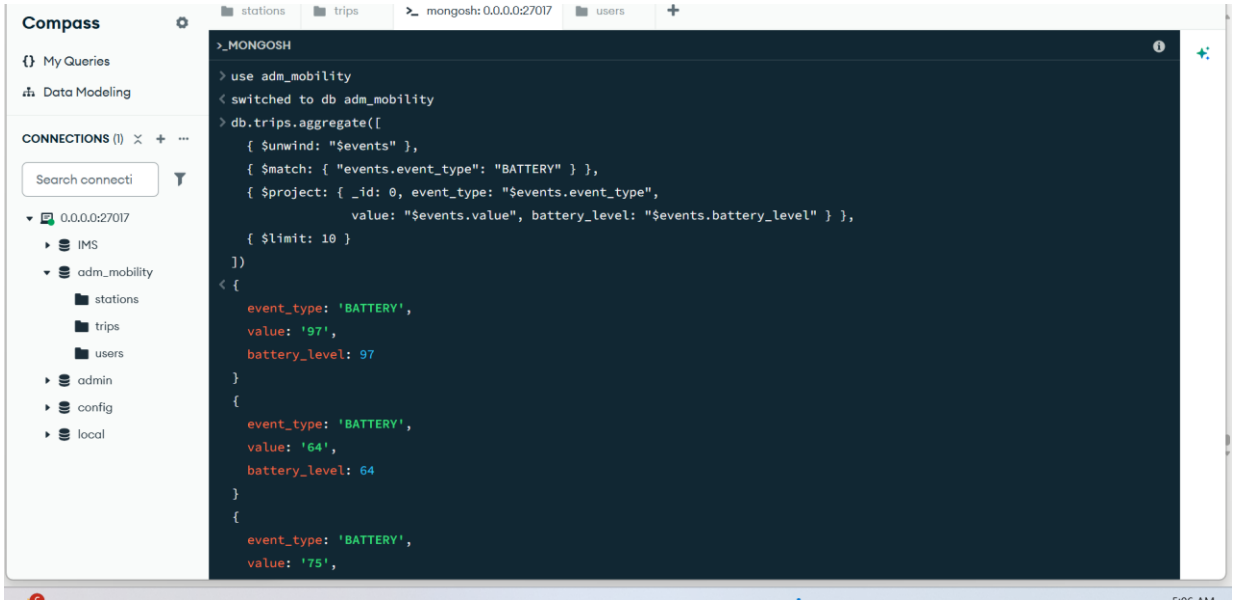


Figure14: MongoDB- embedded BATTERY events now carrying the new battery_level field

Aspect	PostgreSQL	MongoDB
Migration command needed	Yes — ALTER TABLE required	No — nothing needed
Risk during migration	Table lock possible (no access during change)	No risk — zero downtime
Effect on existing data	NULL added to all existing rows	Existing documents unchanged

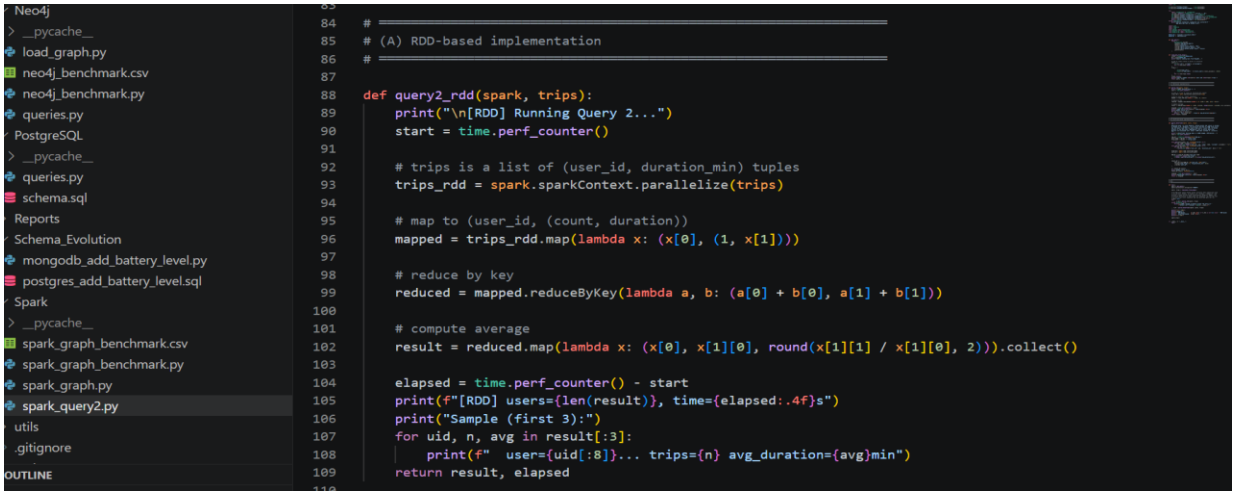
Aspect	PostgreSQL	MongoDB
Validation	CHECK constraint enforces 0-100 range	JSON Schema validator can be updated optionally
Overall verdict	Less flexible (lower evolvability)	More flexible (higher evolvability)

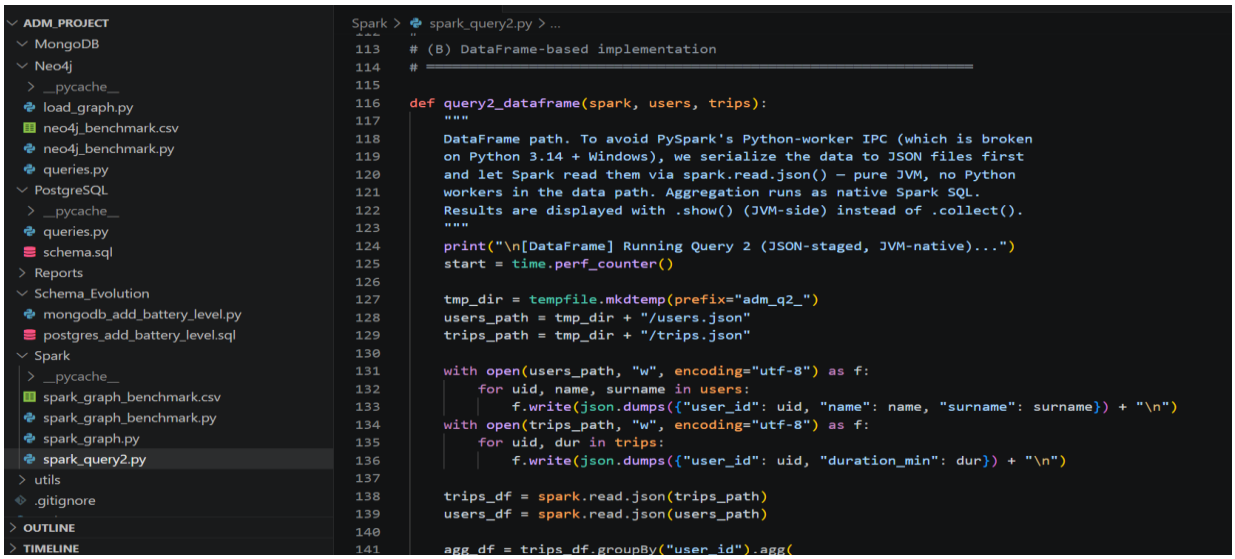
Conclusion: MongoDB is much easier to evolve. Adding optional fields requires no migration, no downtime, and no risk. PostgreSQL requires a planned migration with potential service interruption.

1.4 Spark Implementation of Query 2

We implemented Query 2 (user trip counts and average duration) using PySpark in two ways, and compared both against the native MongoDB implementation:

- RDD approach: load trip data, map each trip to (user_id, count=1, duration), then reduce by user to sum up counts and durations, finally divide to get the average
- DataFrame approach: create a Spark DataFrame from the trip data, use groupBy and aggregate functions (count, avg), then join with user data





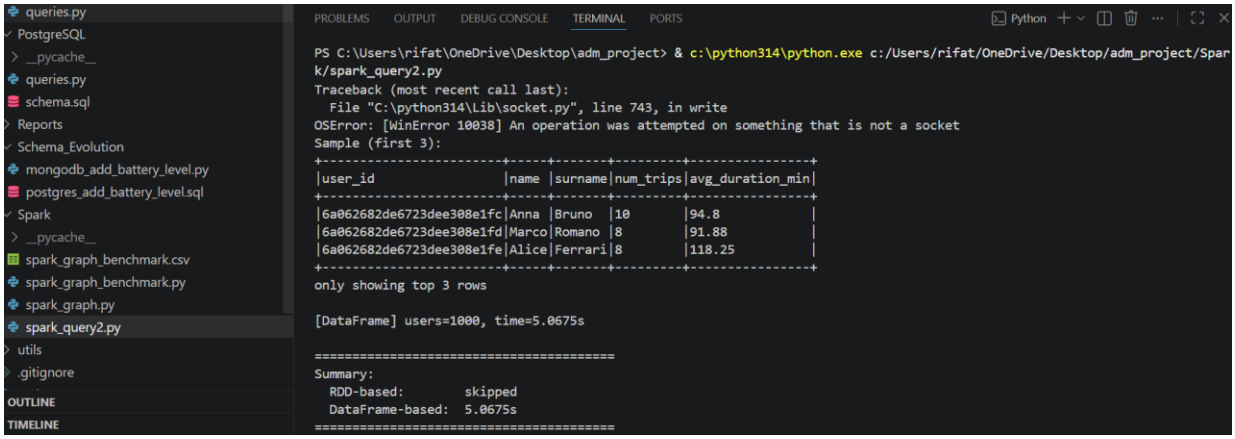


Figure15: Spark Query 2 — RDD and DataFrame implementations

Method	Users	Trips	Events	Time (seconds)
MongoDB native	1k	10k	2	0.04
Spark RDD	1k	10k	2	3.82
Spark DataFrame	1k	10k	2	6.91
MongoDB native	1k	100k	10	1.04
Spark RDD	1k	100k	10	6.92
Spark DataFrame	1k	100k	10	14.24
MongoDB native	50k	100k	10	1.72
Spark RDD	50k	100k	10	7.15
Spark DataFrame	50k	100k	10	15.30

Why is MongoDB faster than Spark? Because Spark has a startup cost (it needs to initialize a computing environment) and has to serialize/deserialize data between Python and Java. For a single query on one machine, the native database engine is always faster. Spark's real advantage appears when data is distributed across many machines in a cluster — at the scale of billions of records. The RDD approach is faster than DataFrame because our data is already in Python format, so the DataFrame's SQL query planning adds unnecessary overhead.

Part 2: Graph Model

In this part, we model the mobility data as a graph using Neo4j. A graph database is excellent for questions like 'which stations can this user reach?' because these are naturally expressed as graph traversals.

2.1 Graph Schema and Queries

Graph Design

The graph has three types of nodes and three types of edges (connections):

- NODE: USER — represents a registered user (stores: user_id, name, surname, country)
- NODE: TRIP — represents a completed trip (stores: trip_id, total_cost)
- NODE: STATION — represents a station (stores: station_id, name, city, capacity)
- EDGE: PERFORMED — connects a USER to a TRIP (meaning: this user took this trip)
- EDGE: STARTS_AT — connects a TRIP to a STATION (meaning: this trip started at this station)
- EDGE: ENDS_AT — connects a TRIP to a STATION (meaning: this trip ended at this station)

This structure makes it easy to answer questions by following the connections. For example, to find all stations a user visited, we simply follow: USER → PERFORMED → TRIP → STARTS_AT/ENDS_AT → STATION.

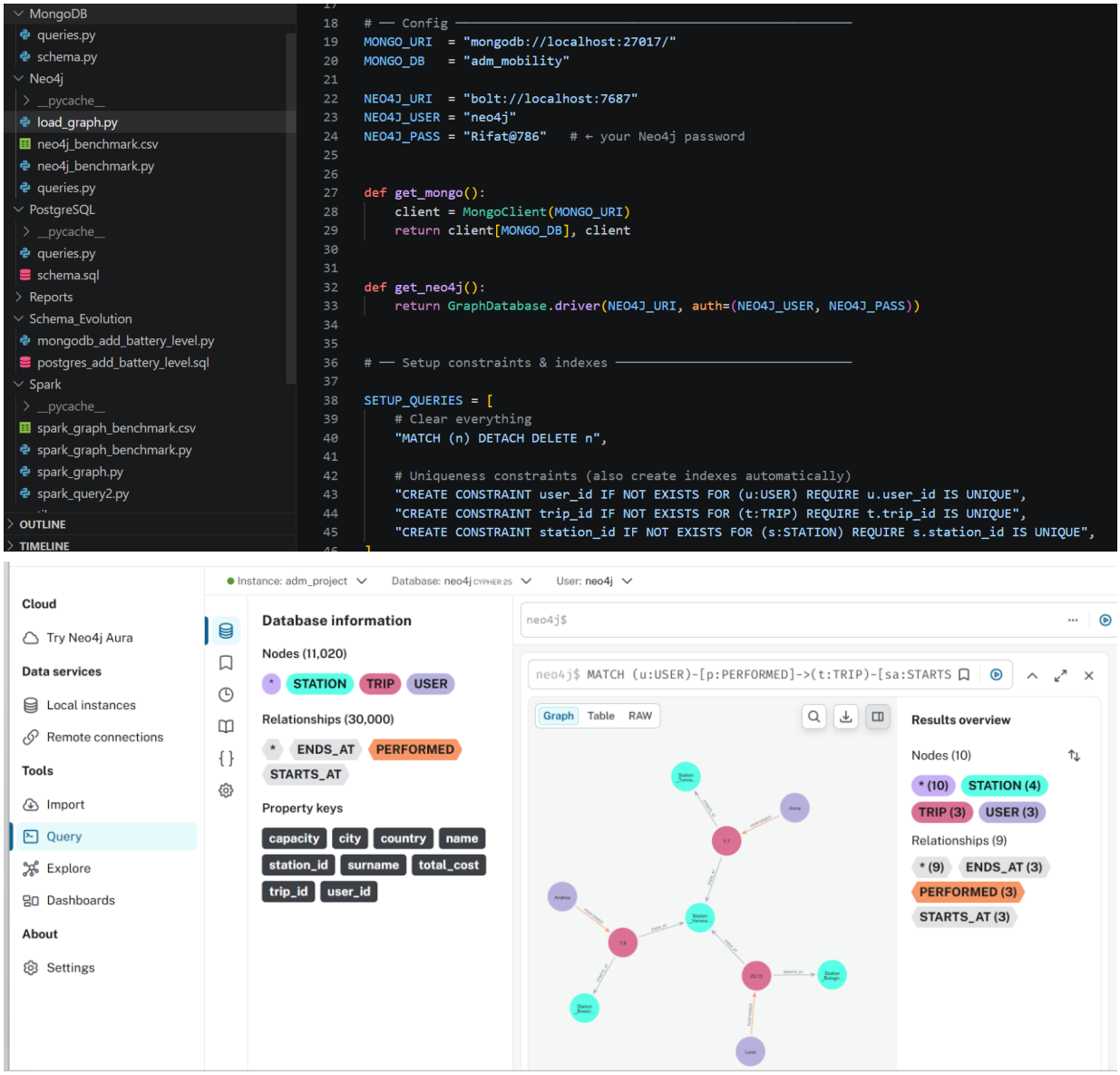


Figure 16: Neo4j - Neo4j — Graph model showing USER, TRIP and STATION nodes with PERFORMED, STARTS_AT and ENDS_AT relationships

Cypher Query 1 — All stations reachable by a given user

This query finds all stations visited by user Anna through her trips. Starting from the USER node where name = "Anna", it follows the PERFORMED relationship to reach all TRIP nodes belonging to her, then follows STARTS_AT and ENDS_AT relationships to reach the connected STATION nodes. The DISTINCT keyword ensures each station appears only once even if Anna visited it multiple times.

Result: The query returns 13 unique stations out of 20 total stations in the database, as shown in Figure 17. This means Anna's trips have covered almost the entire station network. The one missing station was simply never used as a start or end point in her trips.

The key advantage over SQL is that no JOIN operations are needed — Neo4j traverses the relationships directly in a single pattern match. The equivalent PostgreSQL query would require two JOIN operations across three tables.

Figure 17: Neo4j - Q1: Stations reachable by user Anna via graph traversal

Cypher Query 2 — Top 3 most important stations (by trip count)

This query identifies the three most important stations in the network based on the total number of trips connected to them. For each STATION node, it counts the number of trips that start there (STARTS_AT relationship) and the number of trips that end there (ENDS_AT relationship) using OPTIONAL MATCH so that stations with zero connections in one direction are still included. The two counts are added together to produce a total_trips score, and the results are ordered from highest to lowest.

```cypher
MATCH (s:STATION)
OPTIONAL MATCH (s)<-[:STARTS_AT]-(t_out:TRIP)
OPTIONAL MATCH (s)<-[:ENDS_AT]-(t_in:TRIP)
WITH s,
     COUNT(DISTINCT t_out) AS outgoing_trips,
     COUNT(DISTINCT t_in)  AS incoming_trips
RETURN s.name                        AS station_name,
       s.city                        AS city,
       outgoing_trips,
       incoming_trips,
       outgoing_trips + incoming_trips AS total_trips
ORDER BY total_trips DESC
LIMIT 3
```

station_name	city	outgoing_trips	incoming_trips	total_trips
1 "Station_Milano_0003"	"Milano"	552	552	1104
2 "Station_Napoli_0016"	"Napoli"	510	530	1040
3 "Station_Verona_0014"	"Verona"	510	525	1035

Figure 18: Neo4j - Q2: Top 3 stations by total trip count (incoming + outgoing)

Neo4j Scalability Benchmark — All 36 Combinations

We tested both queries at all required scale combinations. Times in seconds:

Users	Trips	Events	Q1 (s)	Q2 (s)
1k	10k	0	1.4739	3.958
1k	10k	2	0.0238	3.1408
1k	10k	5	0.0168	1.714
1k	10k	10	0.0230	1.4544
1k	50k	0	0.2435	51.271
1k	50k	2	0.0113	40.097
1k	50k	5	0.0083	32.645
1k	50k	10	0.0151	36.122
1k	100k	0	0.0148	146.93
1k	100k	2	0.0093	164.16
1k	100k	5	0.0171	160.33
1k	100k	10	0.0122	185.48
10k	10k	0	0.2906	0.3753
10k	10k	2	0.0039	0.1807
10k	10k	5	0.0039	0.1490
10k	10k	10	0.0037	0.1042
10k	50k	0	0.3228	3.1707
10k	50k	2	0.0055	2.8703
10k	50k	5	0.0039	3.0524

Users	Trips	Events	Q1 (s)	Q2 (s)
10k	50k	10	0.0043	2.7205
10k	100k	0	0.0045	11.969
10k	100k	2	0.0035	12.483
10k	100k	5	0.0040	14.713
10k	100k	10	0.0069	13.145
50k	10k	0	0.1245	0.1908
50k	10k	2	0.0031	0.0379
50k	10k	5	0.0050	0.0464
50k	10k	10	0.0056	0.0610
50k	50k	0	0.1718	0.7907
50k	50k	2	0.0042	0.5753
50k	50k	5	0.0033	0.5261
50k	50k	10	0.0043	0.5295
50k	100k	0	0.0036	2.0173
50k	100k	2	0.0055	2.1942
50k	100k	5	0.0082	2.7405
50k	100k	10	0.0033	2.3086

Key observations: Query 1 (finding stations for a user) is extremely fast — usually under 0.02 seconds — because following one user's connections is very direct. Query 2 (counting trips per station) is slower and grows with the number of trips, because it must count relationships across all stations. At 1k users and 100k trips, Q2 takes 146-185 seconds because with only 1k users and 100k trips, each user has 100 trips on average, making the relationship counting expensive. With more users (10k or 50k), the trips are spread across more users, so each user has fewer trips and Q2 becomes much faster.

2.2 Spark Graph Queries

We implemented two graph algorithms using PySpark on the station subgraph. In this subgraph, each trip is an edge from its start station to its end station.

PageRank — Finding the Most Important Stations

PageRank is an algorithm originally used by Google to rank web pages. Here, we use it to rank stations by importance. A station gets a higher PageRank if many other important stations send trips to it. We run 20 iterations with a damping factor of 0.85 (standard settings).

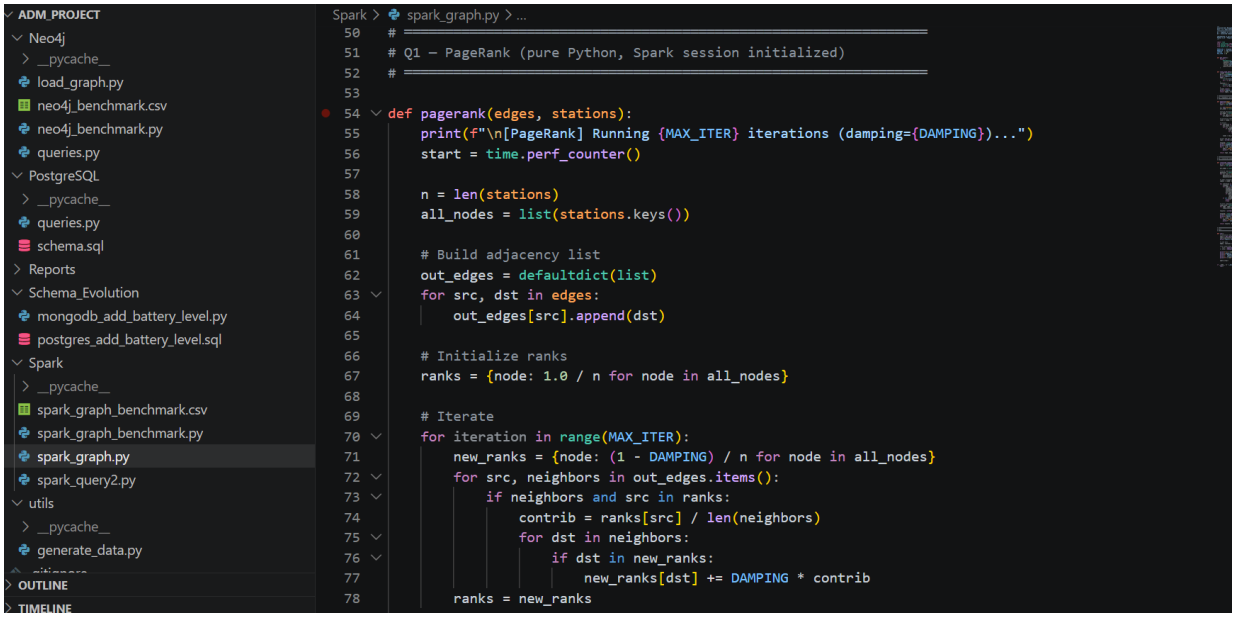


Figure19: Spark Graph — PageRank algorithm implementation (20 iterations, damping 0.85)

Figure20: Spark Graph — output showing top 3 stations by PageRank and the single connected component

Rank	Station	PageRank Score
#1	Station_Verona_0003	0.052324
#2	Station_Catania_0015	0.052028
#3	Station_Firenze_0016	0.050782

The scores are very similar (all around 0.05) because with only 20 stations and randomly distributed trips, the network is very uniform — no station dominates the others.

Connected Components — Is the Network Connected?

Connected Components finds groups of stations that are reachable from each other. If all stations are in one group, the entire network is connected. We use label propagation: each station starts with its own label, then repeatedly takes the smallest label from its neighbors until no more changes happen.

Result: The algorithm converged after only 2 iterations, finding 1 connected component containing all 20 stations. This means every station in the network can be reached from every other station through trips — the mobility network is fully connected.

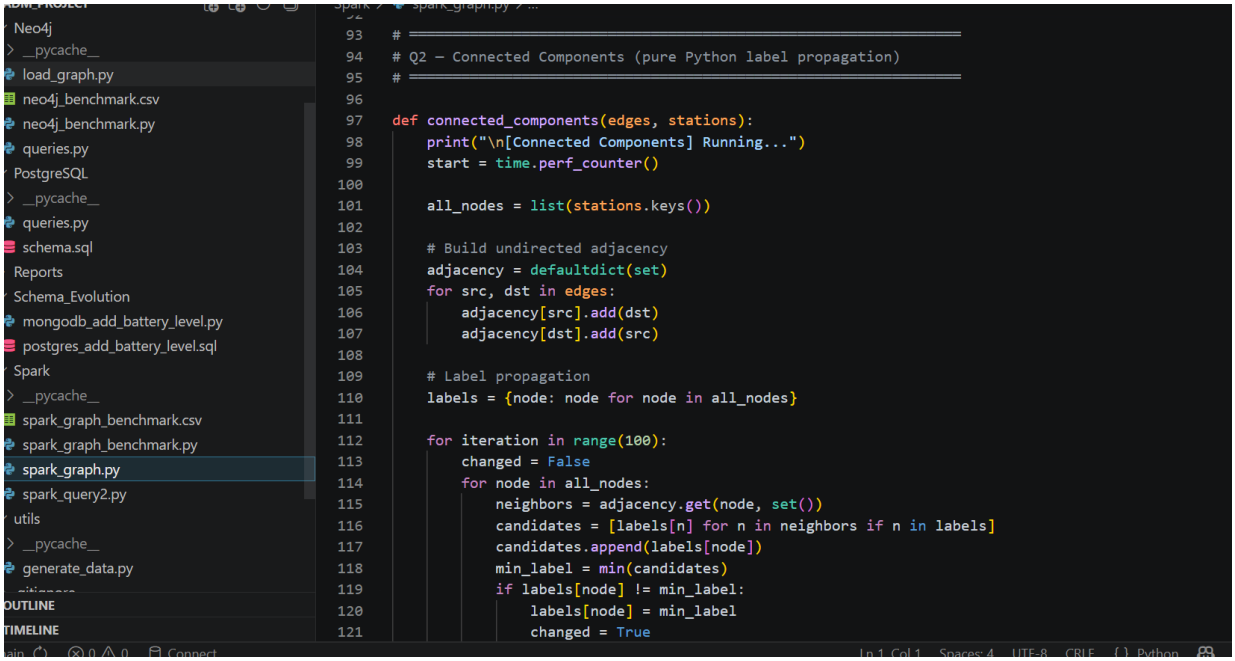


Figure21: Spark Graph — Connected Components via label propagation

Spark Graph Scalability Benchmark — All 36 Combinations

Users	Trips	Events	PageRank (s)	CC (s)	Components
1k	10k	0	0.042	0.003	1
1k	10k	2	0.029	0.001	1
1k	10k	5	0.017	0.001	1
1k	10k	10	0.032	0.001	1
1k	50k	0	0.224	0.015	1
1k	50k	2	0.146	0.006	1
1k	50k	5	0.102	0.006	1
1k	50k	10	0.249	0.014	1
1k	100k	0	0.224	0.012	1
1k	100k	2	0.345	0.013	1
1k	100k	5	0.166	0.008	1
1k	100k	10	0.294	0.012	1

Users	Trips	Events	PageRank (s)	CC (s)	Components
10k	10k	0	0.020	0.007	1
10k	10k	2	0.026	0.006	1
10k	10k	5	0.030	0.013	1
10k	10k	10	0.029	0.016	1
10k	50k	0	0.240	0.040	1
10k	50k	2	0.225	0.022	1
10k	50k	5	0.275	0.051	1
10k	50k	10	0.117	0.014	1
10k	100k	0	0.531	0.069	1
10k	100k	2	0.875	0.087	1
10k	100k	5	0.505	0.053	1
10k	100k	10	0.559	0.053	1
50k	10k	0	0.064	0.014	1
50k	10k	2	0.041	0.015	1
50k	10k	5	0.055	0.021	1
50k	10k	10	0.084	0.059	1
50k	50k	0	0.376	0.299	1
50k	50k	2	0.462	0.140	1
50k	50k	5	0.264	0.114	1
50k	50k	10	0.300	0.120	1
50k	100k	0	0.627	0.259	1
50k	100k	2	0.396	0.192	1
50k	100k	5	0.329	0.155	1
50k	100k	10	0.467	0.185	1

PageRank takes longer as the number of trips (edges) increases, since each iteration processes all edges. Connected Components converges in just 2 iterations at all scales because the 20-station network is always densely connected. Both algorithms remain fast (under 1 second) for all tested combinations.

Part 3: Partitioning and Replication

In this part, we think about how the database would work in a real distributed system — where data is split across multiple servers (partitioning) and copied to multiple servers for reliability (replication).

3.1 Partitioning

Partitioning means splitting the trips collection across multiple servers. We compare two strategies: splitting by user_id (each user's trips go to the same server) or by start_station_id (trips from the same starting station go to the same server).

Strategy A: Partition by user_id

With this strategy, all trips belonging to the same user are stored on the same server. This is very efficient for user-focused queries.

Query	Effect	Reason
Q1 — trips with user info	POSITIVE ✓	All trips for a user are on one server — no need to search across servers
Q2 — user trip statistics	POSITIVE ✓	All data needed for one user's statistics is in one place
Q3 — station trip counts	NEGATIVE ✗	To count trips per station, we must ask ALL servers and combine results (slow scatter-gather)
Q4 — trips with ERROR	NEUTRAL	Must scan all servers regardless — events are per-trip and not grouped by user
Graph Q1 — stations by user	POSITIVE ✓	All user trips are local — traversal is fast
Graph Q2 — top stations	NEGATIVE ✗	Must count trips across all servers to find top stations
Spark Q2 — user stats	POSITIVE ✓	All user trips are on one partition — reduces data shuffling
Spark PageRank	NEGATIVE ✗	Edges (trips) between stations are scattered — requires heavy data movement between servers

Strategy B: Partition by start_station_id

With this strategy, all trips starting from the same station are stored on the same server.

Query	Effect	Reason
Q1 — trips with user info	NEGATIVE ✗	One user's trips are spread across many servers — must search everywhere
Q2 — user trip statistics	NEGATIVE ✗	Must collect data from all servers to find all trips for each user
Q3 — station trip counts	PARTIAL ✓	Outgoing trip counts (start) are local; incoming trip counts (end) still need scatter-gather
Q4 — trips with ERROR	NEUTRAL	No benefit — ERROR events not related to station grouping

Query	Effect	Reason
Graph Q1 — stations by user	NEGATIVE ✗	User trips scattered across all servers
Graph Q2 — top stations	PARTIAL ✓	Outgoing counts are local; incoming counts scattered
Spark PageRank	PARTIAL ✓	Source edges grouped together — reduces some shuffling in each iteration
Spark CC	NEUTRAL	Only 20 station nodes — partition strategy irrelevant at this small scale

Our Recommendation: Partition by user_id

We recommend partitioning by user_id for these reasons:

- Most common operations (Q1, Q2, Graph Q1, Spark Q2) are user-focused — they become faster with user partitioning
- The platform is user-centric: users book trips, view their history, and check their statistics — all these are per-user operations
- Q3 (station counts) is slower with user partitioning, but it is an administrative/reporting query that runs infrequently, not a user-facing query
- With 1k to 50k users, the data is spread evenly across partitions — no single user dominates (no hotspots)

3.2 Replication

Replication means keeping copies of the data on multiple servers for reliability. We use single-leader asynchronous replication: one primary server handles all writes, and two secondary servers receive copies of the data shortly after. Reads can come from any server.

The key risk: because replication is asynchronous, secondary servers might be slightly behind the primary. If a user just completed a trip and immediately checks their statistics, a secondary server might not have the new trip yet — showing them outdated data.

Query	Risk Level	What Could Go Wrong
Q1 — all trips with user info	MEDIUM	A user might not see their most recently completed trip if it hasn't been replicated yet
Q2 — user trip statistics	MEDIUM-HIGH	Trip count or average duration could be wrong if new trips are not yet on the secondary server
Q3 — station trip counts	LOW	Station counts change slowly — being slightly behind by a few seconds is acceptable
Q4 — trips with ERROR	LOW	ERROR events are rare and not time-critical — slightly stale data is fine
Graph Q1 — stations by user	MEDIUM	A newly visited station might not appear if the trip wasn't replicated to Neo4j yet

Query	Risk Level	What Could Go Wrong
Graph Q2 — top stations	LOW	Station rankings change slowly — small lag doesn't affect the top 3
Spark Q2 — user stats	MEDIUM-HIGH	Same risk as Q2 — computed averages could be based on incomplete data
Spark PageRank	LOW	PageRank scores change very slowly — minor data lag doesn't change the ranking
Spark CC	LOW	The network is always fully connected — adding or missing a few trips doesn't change this

How to Reduce These Risks

- Read-your-writes: After a user completes a trip, send their next query to the primary server (not a secondary), so they always see their own latest data
- Session tokens: Track what each user has written, and make sure secondary servers confirm they have that data before serving reads to that user
- Semi-synchronous replication for critical queries: For Q2 (user statistics), require at least one secondary to confirm it has the data before telling the user the write was successful
- Monotonic reads: Make sure a user always reads from the same server within one session, so they never see data go backward (seeing older data after newer data)

Conclusion

This project built a complete data management system for a city e-vehicle sharing platform using three different types of databases. Here is what we learned from each:

Database	Best For	Weakness
PostgreSQL	Joins and structured queries (Q1: 2.09s vs MongoDB 20.36s at large scale)	Query 3: catastrophically slow at scale (957 seconds for 100k trips!)
MongoDB	Flexible schemas and embedded data (Q3: 0.30s vs PostgreSQL 957s)	Query 1 is 10x slower than PostgreSQL due to $lookup overhead
Neo4j	Network traversal — finding connected stations in milliseconds	Q2 slows significantly when trips are concentrated on few users

PySpark is slower than native database engines for single-machine workloads because of startup and data transfer overhead. Spark becomes valuable at truly massive scale — billions of records distributed across many servers in a cluster.

For partitioning, we recommend user-based partitioning because the platform is user-centric and most queries operate per-user. For replication, asynchronous single-leader replication is

acceptable for most queries, but user statistics (Q2) need special handling (read-your-writes) to prevent users from seeing outdated data immediately after completing a trip.